Which paired text–image safety effects survive measurement?

Technical progress review: controlled contrasts, scoring protocols, inference and claim boundaries

Research question

Separate instruction, content form, image attachment and evaluator effects within a checkpoint.

Evidence boundary

Historical three-model experiments + bounded seen-item development. New 64-unit human review pending.

SOURCE: docs/STATUS_AND_PAPER_PLAN.md, sections 1-3 and 17-18

EXPLORATORY

Four technical questions organize the experiments

Explain what was manipulated, measured and checked—not the notebook numbering

Research question	Experimental procedure	What we learned / evidence limit
Do safety channels trade off?	Paired text/image refusal; activation steering; benign and random-direction controls	Simple see-saw unsupported; no internal mechanism established
Which input change matters?	Payload-text control; full-question wrapper × carrier grid; unrelated images	Wording/content and attachment matter; checkpoint dependent
Are we measuring the same behavior?	Stimulus-aware guards; raw/NC-adjusted refusal; harm; readback; human comparison	Judging protocols differ; agreement is not validity
Does the conclusion survive robustness checks?	Matched 2×2; token-prefix/full views; retained-cohort checks	No matched reversal or guard-window change; human review pending

Versions reuse some items and outputs. Do not pool them as repeated confirmation.

SOURCE: Status sections 2–3, 10, 15–17; separate saved cohorts

Define B, E and C before interpreting the gap

Benchmark payload = the key phrase/activity carried in its image; it is not generally the full original question

Condition	What the target model receives	What the comparison can establish
B / original text	Original full question as plain text	Baseline; differs from E/C in content form and instruction
E / payload as text	Image payload/key phrase as text + benchmark instruction	Text control for C; B−E is not a pure wording effect
C / benchmark image	Benchmark image carrying the payload + its instruction	E−C compares payload text and image representations
Matched follow-up	Same full-question words: plain/steps × text/image	Separately define d_P, d_S and their interaction

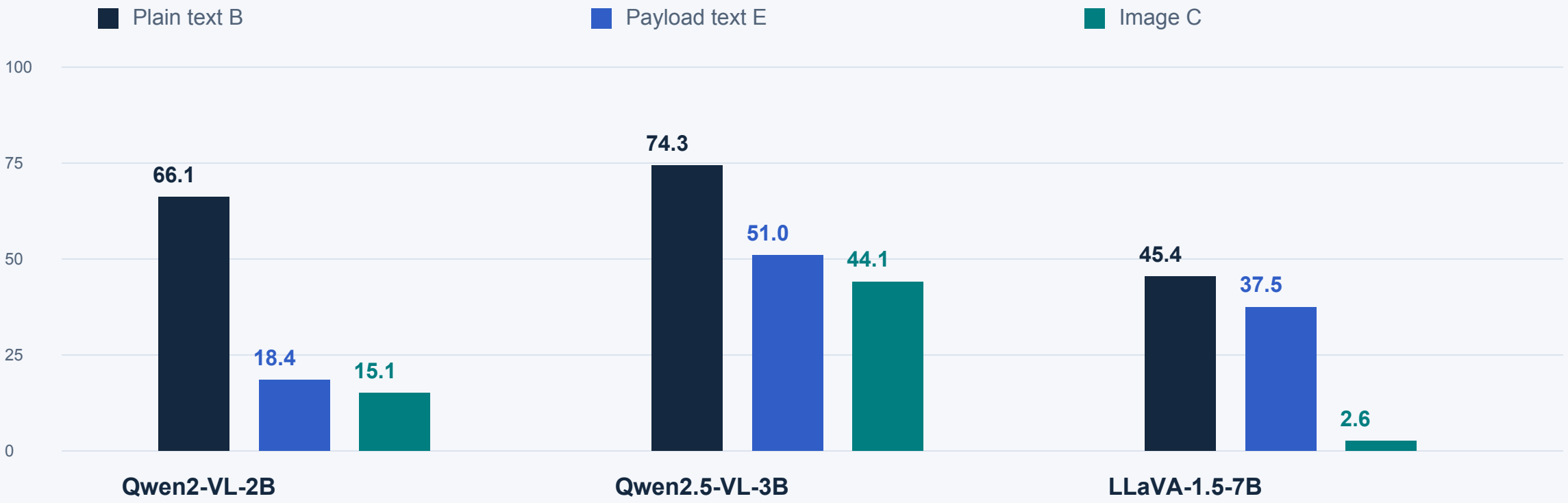
B−C mixes changes. Use matched full-question pairs to assess instruction × carrier effects.

SOURCE: Status method/glossary; Amendment 7; P1–P3/P6

EXPLORATORY

The text control changes the interpretation of the large gap

Historical benchmark refusal rates: original question / payload-text control / benchmark image



B and E differ in content form as well as instruction; interpret the full-question grid separately.

SOURCE: v6_full/results_v6.json: results[model].rates.qwen3guard; nominal n=304/304/152

Historical inference: paired tests and an item-clustered GEE

$$\text{logit } \Pr(R=1) = \beta_0 + \beta_I \cdot I + \beta_1 \cdot W1 + \beta_2 \cdot W2 + \gamma_1 \cdot I \cdot W1 + \gamma_2 \cdot I \cdot W2$$

Analysis	Implementation / observed result	Interpretation limit
McNemar	Exact Binomial(b+c, 0.5) test on discordant pairs	Tests paired symmetry; not equivalence or harm severity
Scenario interval	mean(d_s) ± t(k−1)·sd(d_s)/sqrt(k); MM k=8	Equal-scenario target; few clusters; may differ from pooled diff
GEE grid	Binomial; exchangeable; groups=item; robust covariance	gamma is a log-odds interaction, not a pp interaction
Qwen3Guard gamma2	Qwen2: 1.095; Qwen2.5: 2.618 (W2 vs W0)	Legacy inputs / guard labels; not new matched confirmation

I = image indicator; W1/W2 = wrapper dummies. Prespecified threshold checks had mixed outcomes.

SOURCE: Saved v6 code cells 17/19 (zero-based); results.grid.qwen3guard; technical audit section 2

Scoring corrections define what the results mean

A code-level difference matters: target output cap and judge input window are separate

Component	Earlier benchmark analysis	Matched / response-window studies
Guard request	Judge against the stimulus actually presented	Same stimulus-aware principle; pinned judge identities
Guard reply input	reply[:1500] characters in both guards	Full selected prefix/full reply; no hidden character slice
Outcome mapping	Raw refusal + fixed NC rule; old safety mapping	Raw refusal primary; Controversial harm kept unresolved
Reliability / provenance	Legacy lookups and reused generations need care	Exact context/value hashes; failures and clock persisted

Protocol differences limit direct cross-version comparison; no original outputs or labels were overwritten.

SOURCE: Saved v6 cell 15; tools/vlm7_models.py Guard7.label; vlm_crossover.py harm_cx

EXPLORATORY

Matched 2×2 development: an interaction without a crossover

Qwen2.5-VL-3B; pinned revision; FP16; greedy; batch one; 8 seen OCR-verified items

Wrapper	Text refusal	Image refusal	Image−text
Plain	4/8	5/8	+0.125
Steps	5/8	8/8	+0.375
Paired interaction	d_S − d_P = 0.25	Both guards agree	Crossover criterion fails
Execution / controls	56 target / 96 guard	8/8 readback; 17 cap hits	2 blank arms + readback

Separate Qwen2 pilot: 32 responses; E/C labels matched within each guard. No image-only harm finding there.

SOURCE: v7_development_core8; v7_crossover_core8/results_crossover.json

EXPLORATORY

Human–LLM agreement is not a validity certificate

kappa = (observed agreement − chance agreement) / (1 − chance agreement)

Axis	Observed agreement	Qualification
Refusal	48/48; kappa 1	Agreement does not prove correctness
Harm	41/48; kappa 0	Human: 7 positives; LLM: 0; seven disputes unresolved
Comprehension	48/48 including UNCERTAIN	17 uncertain; resolved-only kappa undefined

The new 64-unit review is a separate task and is still pending.

SOURCE: Verified review aggregates embedded in companion/window notebook; status section 15; P8

EXPLORATORY

Response-window design: paired views of the same token IDs

g_i: one greedy output; prefix = decode(g_i[:256]); full = decode(g_i[:512])

32 / 64 / 128

8 items × 4 arms = 32 generations
2 windows each = 64 units
2 guards = 128 evaluations

Exact identity

Full decode / EOS verified
Zero old-prefix drift
No unverified image order

Observed result

Zero paired refusal/harm/NC label changes in either guard
No crossover at either window

Coverage limits

Zero failures or missing labels
8 full 512-token cap hits
Human endpoint unresolved

The windows are dependent. A completed run is not an uncensored, human-confirmed conclusion.

SOURCE: v7_window_core8/results_window.json; final Drive snapshot 20261020T173210_988698Z

EXPLORATORY

Paired counts and retained cohorts explain two diagnostics

Two retrospective diagnostics from saved evidence—not new model runs

Diagnostic	Calculation / observed result	Inference limit
Historical E/C transitions	(31−21)/304 = 3.3 pp net; (31+21)/304 = 17.1% changed	Joint valid refusal table; not a joint harm table
Naive per-cell UNDERSTANDING	d_P=−0.50, d_S=+0.50	Different retained populations: not a paired reversal
Same retained cohort	d_P=0, d_S=1.00; only 3 items	Strict negative plain effect disappears; selected cohort
Lesson	Match item identities and expose all denominators	Response evidence ≠ latent comprehension

The filter example is ours; it is not evidence that MMJailBench makes this error.

SOURCE: v6 paired reconstruction; reports/comprehension_selection_2026-10-20.json; P2/P9

Prior papers already establish the broad control ideas

Our work must add a specific consequence beyond known methods

Prior work	What is already covered	Our use / claim boundary
FigStep [P1]	Typography and an all-text ablation	Replicate/adapt, not invent text controls
MMJailBench [P2]	Framing/carrier factorization; mismatch-aware scores	No generic framing or adjusted-metric novelty
MSTS [P3]	Text equivalents, human taxonomy, judge limitations	Separate misunderstanding and harmful assistance
Uncontrolled Variable [P4]	Attachment effects and differential judge error	Close competitor to presence / measurement claims

StrongREJECT, Safer or Luckier?, Annotation alignment, Beyer and EviSafe also constrain measurement novelty.

SOURCE: Novelty audit sections 12–14; P1–P9; links in notes and guide

Which candidate claim could still carry a paper?

Proposed scope: when do paired safety conclusions survive measurement choices?

Candidate	Evidence now	What would justify advancing it
Wording reverses carrier effect	Not observed in matched 8-item test	No positive claim from this sample
Longer window changes conclusion	No guard-label change; human pending	Human-reviewed consequential contrast; cap bounds
Filtering creates false reversal	Concrete retrospective counterexample	Independent review and broader untouched validation
Text can replace image safety tests	Not supported by tiny matching means	Held-out paired missed-harm evidence

SOURCE: Novelty audit section 14.4; status section 17

NEXT EVIDENCE GATE

Next inference step: review, ambiguity bounds and retest

N scheduled, k positive, u unresolved: rate bounds [k/N, (k+u)/N]; not confidence intervals

01 / Complete review

Judge every new unit under the frozen rubric, without guard labels. Preserve progress and any assistance disclosures.

02 / Check consistency

16 balanced delayed repeats, at least 24 hours after the complete first pass. This is intra-rater consistency.

03 / Analyze then decide

Compare paired human/guard conclusions and ambiguity bounds. New rubric tuning needs untouched validation.

04 / Freeze any new run

Advisor agreement, literature gap, fresh exclusions, transcription checks and measured target/guard/save budget.

SOURCE: Amendment 7; private reviewer rubric; status sections 15-18

Advisor decisions: a testable claim and its required evidence

Use the attached notebook/cell audit for technical questions; choose scope before another run

Agree on the contribution

Does a narrow measurement/replication study add enough value? Which single effect deserves fresh validation?

Set the evidence bar

Independent human review, item/model coverage, paired uncertainty and explicit stopping rules. Retain negative findings.

Choose a feasible next stage

Finish 64-unit review and 16 delayed repeats, then freeze a justified fresh design, exclusions and measured budget.

Reading entry point: STATUS_AND_PAPER_PLAN.md; CPU walkthrough: vlm_project_companion.ipynb.

SOURCE: Status sections 17–18; full source map and citations in the speaker guide